

2025

SUSTAINABILITY IMPACT REPORT 2025

Building AI Infrastructure for a Livable Planet

A comprehensive assessment of Harch Corp's environmental impact — 89% lower carbon intensity, 81.5% renewable energy, zero water cooling at three hubs, and a measurable path to net-zero operations by 2030.

47

GCO₂/KWH AVG

81.5%

RENEWABLE ENERGY

<1.15

PUE AVERAGE

89%

BELOW INDUSTRY AVG

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HARCH.

SOVEREIGN. SUSTAINABLE. AI INFRASTRUCTURE.

Executive Summary

Sustainability is not an afterthought at Harch Corp; it is the foundational principle upon which every infrastructure decision is made. As the global AI industry confronts an escalating carbon crisis -- with data center emissions projected to exceed 3.5% of global electricity consumption by 2030 -- Harch has engineered a fundamentally different approach to AI compute. Rather than retrofitting sustainability onto conventional infrastructure, we have built our platform from the ground up with carbon awareness at its core, achieving an average grid carbon intensity of approximately 47 gCO₂/kWh across all five hubs, which is 89% below the industry average of approximately 450 gCO₂/kWh.

This report presents a comprehensive account of Harch Corp's environmental impact across the full lifecycle of its AI infrastructure, from the embodied carbon of hardware procurement through operational emissions and end-of-life stewardship. We document the measurable outcomes of our carbon-aware scheduling engine, which routes workloads in real time to the hub with the lowest carbon intensity at any given hour, reducing average operational carbon intensity by approximately 60% compared to static workload placement. With 81.5% average renewable energy penetration, a PUE below 1.15 versus the industry average of 1.4-1.6, and zero water usage for cooling at three of our five hubs, Harch demonstrates that high-performance AI compute and environmental responsibility are not opposing forces but complementary objectives.

Our five GPU hubs across the Kingdom of Morocco leverage the country's exceptional renewable energy resources -- 2,800 kWh/m₂/year solar irradiance, Atlantic wind corridors averaging 9.7 m/s at Dakhla, and a national renewable energy strategy targeting 52% by 2030 -- to deliver compute infrastructure that is simultaneously faster, cheaper, and dramatically cleaner than hyperscaler alternatives. This report is an invitation to our customers, partners, and the broader industry to examine our data, challenge our assumptions, and join us in building AI infrastructure that the planet can sustain.

The Carbon Crisis in AI

AI's Growing Carbon Footprint

The carbon footprint of artificial intelligence has emerged as one of the most pressing environmental challenges of the digital age. Training a single large language model can emit over 300 tonnes of CO₂, equivalent to the lifetime emissions of five average automobiles. The International Energy Agency projects that global data center electricity consumption will double between 2022 and 2026, driven primarily by AI workloads. This trajectory is unsustainable: if current growth rates continue, AI infrastructure alone could consume 4-6% of global electricity generation by 2030, generating carbon emissions that would effectively negate the climate benefits of many renewable energy deployments.

The root cause lies in the convergence of three trends. First, the computational demands of state-of-the-art AI models have been doubling approximately every 3.4 months, far outpacing the efficiency gains predicted by Moore's Law. Second, the geographic distribution of GPU infrastructure remains heavily concentrated in regions with fossil-fuel-intensive grids: the average carbon intensity of electricity in major cloud computing regions is approximately 450 gCO₂/kWh, with some regions exceeding 600 gCO₂/kWh. Third, the industry's approach to sustainability has been predominantly reactive, focused on purchasing carbon offsets rather than fundamentally redesigning infrastructure to minimize emissions at the source.

Moore's Law for Compute vs. Climate

While computational performance has followed an exponential curve, the carbon efficiency of compute has improved only linearly. The result is an ever-widening gap between what AI can do and what the planet can sustain. A 2023 study by the University of Massachusetts Amherst found that the carbon emissions from training a single state-of-the-art model increased by a factor of 12 between 2019 and 2023, even as model performance on standard benchmarks improved by only 40%. This asymmetry reveals a fundamental truth: efficiency gains within existing infrastructure paradigms cannot keep pace with the growth in compute demand. A structural change in where and how AI compute is provisioned is required.

The Regulatory Imperative

Governments and regulators are responding to this crisis with unprecedented urgency. The EU Corporate Sustainability Reporting Directive (CSRD), which took effect in January 2024, mandates comprehensive carbon disclosure for all large companies operating in the European Union, including the emissions associated with purchased compute services. The EU AI Act further requires that high-risk AI systems be developed with environmental sustainability as a design principle, not a post-hoc consideration. California's SB 253 and SB 261 impose similar disclosure requirements on companies doing business in the state. For organizations that rely on AI compute, these regulations transform the carbon intensity of their infrastructure from a voluntary metric into a mandatory disclosure with legal and financial consequences.

Metric	Current Value	Trend	Source
Global DC electricity share	1.5-2%	Doubling by 2026	IEA 2024
Avg cloud carbon intensity	~450 gCO ₂ /kWh	Flat/slow decline	Ember Climate
Training emissions (70B model)	~300 tCO ₂	+12x since 2019	UMass Amherst
AI compute demand growth	2x per 3.4 months	Accelerating	Epoch AI
CSRD-covered companies	~50,000 EU firms	Mandatory from 2024	EU Commission

Table 1: Key metrics illustrating the carbon crisis in AI compute.

Morocco's Renewable Advantage

Exceptional Solar Irradiance

The Kingdom of Morocco possesses one of the most favorable solar energy environments on Earth, with an average solar irradiance of 2,800 kWh/m₂/year, approximately 40% above the best regions in Southern Europe and more than double the irradiance available in Northern Europe. This extraordinary solar resource is not merely a statistical curiosity; it translates directly into lower-cost, lower-carbon electricity that fundamentally changes the economics and environmental impact of energy-intensive operations such as AI compute. The Noor Ouarzazate Solar Complex, the largest concentrated solar power facility in the world with over 580 MW of installed capacity, sits adjacent to Harch's flagship GPU hub and demonstrates the country's commitment to utility-scale solar deployment.

Morocco's National Energy Strategy targets 52% renewable energy by 2030 and has already achieved approximately 42% installed renewable capacity. The country has attracted over \$13 billion in renewable energy investment since 2009, with major projects including the Noor solar complex, the 850 MW Tarfaya wind farm, and numerous distributed solar installations. This investment trajectory ensures that the renewable energy available to Harch's infrastructure will continue to increase, progressively lowering the carbon intensity of each hub over time.

Atlantic Wind Corridors

Morocco's Atlantic coastline benefits from persistent trade winds that create some of the most consistent and powerful wind resources in the world. The Dakhla region records average wind speeds of 9.7 m/s, placing it among the top wind energy sites globally. The Tarfaya wind farm, one of the largest in Africa, operates with a capacity factor exceeding 45%, far above the global average of approximately 30% for onshore wind. These wind resources complement solar generation by providing peak output during evening and nighttime hours when solar generation declines, creating a natural hybrid energy profile that delivers more consistent renewable supply than either resource alone.

The energy cost advantages are equally compelling. Dakhla's wind-generated electricity costs approximately \$0.018/kWh, which is 72% below the European average of \$0.064/kWh. This cost advantage is secured through 15-year Power Purchase Agreements (PPAs) that provide long-term price stability, insulating Harch's operating costs from the volatility that plagues fossil-fuel-dependent grids. For Harch's customers, this translates into GPU compute costs that are structurally lower than those of hyperscaler competitors, while simultaneously delivering a dramatically lower carbon footprint.

Geographic Proximity to Europe

Morocco's geographic position at the northwestern corner of Africa, separated from Europe by just 14 km at the Strait of Gibraltar, provides a unique advantage: access to world-class renewable energy resources with network latency to European financial and technology centers that is comparable to intra-European paths. The 8 ms

round-trip latency from Tanger to Madrid, and approximately 22 ms to Frankfurt, enables Harch to serve latency-sensitive European inference workloads from infrastructure powered predominantly by renewable energy -- a combination that no European-based data center can match on either dimension simultaneously. This geographic proximity, combined with four submarine cable systems providing redundant multi-Tbps connectivity, makes Morocco the ideal location for sustainable AI infrastructure serving European, African, and Middle Eastern markets.

Resource	Morocco	Southern Europe	Northern Europe	Advantage
Solar irradiance (kWh/m ² /yr)	2,800	1,900-2,000	900-1,200	+40% vs S. Europe
Wind speed, Dakhla (m/s)	9.7	6-7 (best sites)	5-6	+45% vs S. Europe
Wind capacity factor	45%+	25-30%	25-35%	+50% vs S. Europe
Energy cost, wind (\$/kWh)	\$0.018	\$0.04-0.05	\$0.05-0.08	-72% vs EU avg
Renewable target (2030)	52%	40% (EU avg)	50-60%	Ambitious pathway

Table 2: Morocco's renewable energy resources compared to European benchmarks.

Carbon-Aware Scheduling: How It Works

Real-Time Grid Monitoring

The Harch Platform's carbon-aware scheduling engine, HarchOS, ingests real-time grid carbon intensity data from multiple sources to maintain a continuously updated picture of the carbon cost of electricity at each hub location. Primary data feeds include the Moroccan grid operator ONEE (Office National de l'Electricite et de l'Eau Potable) real-time generation mix API, the European ENTSO-E Transparency Platform for cross-border electricity flows, and on-site solar irradiance and wind speed sensors at each hub that predict renewable generation output for the next 24-48 hours with high accuracy. These data streams are fused through a machine learning model that produces a carbon intensity forecast for each hub at 15-minute intervals, enabling forward-looking scheduling decisions rather than merely reactive ones.

The monitoring system tracks not only current carbon intensity but also the marginal carbon intensity, which represents the emissions associated with the next unit of electricity generated. This distinction is critical because adding GPU load to a grid that is currently generating surplus renewable energy has a very different carbon impact than adding the same load when the grid is relying on fossil fuel peaking plants. HarchOS optimizes for marginal carbon intensity, ensuring that workload placement decisions reflect the true incremental carbon cost of compute, not merely the average.

Workload Routing Algorithm

When a workload is submitted to the Harch Platform, HarchOS evaluates a multi-objective cost function that incorporates five dimensions: current marginal carbon intensity at each hub, available GPU capacity and hardware configuration, data residency constraints mandated by the customer's compliance requirements, network latency to the end user or data source, and energy cost at each hub. The relative weights of these dimensions are configurable per workload, allowing customers to express their own priorities. A sustainability-focused research institution might weight carbon intensity at 60%, while a financial trading platform might weight latency at 70%. HarchOS resolves these multi-objective constraints through a Pareto-optimal scheduling algorithm that identifies placement options satisfying all hard constraints before selecting the option that minimizes the weighted cost function.

Deferral for Batch Workloads

For batch training workloads that have flexible scheduling windows, HarchOS implements an intelligent deferral mechanism that shifts execution to periods of peak renewable generation. Historical grid data shows that solar generation in the Ouarzazate region peaks between 10:00 and 15:00 local time, during which carbon intensity drops to as low as 8 gCO₂/kWh. By aligning compute-intensive workloads with these windows, the platform achieves a 15-30% reduction in operational carbon emissions compared to immediate scheduling. The deferral mechanism is constrained by the customer's deadline: if the project timeline does not permit waiting for the next renewable peak, the workload is executed immediately at the lowest-carbon hub with available

capacity.

Measurable Outcomes

The impact of carbon-aware scheduling is quantifiable and substantial. Across all workloads executed on the Harch Platform, carbon-aware scheduling reduces the average operational carbon intensity by approximately 60% compared to static workload placement (where each customer is assigned a fixed hub). This figure accounts for the full range of workload types: batch training jobs that benefit from deferral achieve reductions of 60-75%, while latency-sensitive inference workloads that cannot be deferred still achieve reductions of 30-45% through geographic routing to the lowest-carbon hub within latency budget. The 200 MWh battery storage system deployed across the hub network further enhances these outcomes by providing grid transition backup, ensuring that renewable energy supply can be smoothed across intermittent generation periods without resorting to fossil-fuel grid power.

Workload Type	Static Placement	Carbon-Aware	Reduction	Mechanism
Batch training	~120 gCO ₂ /kWh	~36 gCO ₂ /kWh	60-75%	Temporal deferral
Fine-tuning	~100 gCO ₂ /kWh	~45 gCO ₂ /kWh	50-60%	Hub selection + deferral
Real-time inference	~80 gCO ₂ /kWh	~50 gCO ₂ /kWh	30-45%	Geographic routing
MLOps / CI pipelines	~90 gCO ₂ /kWh	~40 gCO ₂ /kWh	50-65%	Off-peak scheduling
Platform average	~82 gCO ₂ /kWh	~47 gCO ₂ /kWh	~60%	Combined approach

Table 3: Carbon intensity reduction by workload type under carbon-aware scheduling.

Five Hubs: Carbon Profile

Each of the five Harch GPU hubs has a distinct carbon profile shaped by its geographic location, available renewable energy sources, grid interconnection, and cooling architecture. This section presents a detailed account of the carbon intensity, renewable energy sources, power usage effectiveness, and free cooling potential of each hub, enabling customers and auditors to make informed decisions about workload placement and verify Harch's environmental claims against measurable data.

Ouarzazate: Solar-Dominant Profile

The Ouarzazate hub achieves 97.2% renewable energy through a combination of direct solar generation from the adjacent Noor Ouarzazate Solar Complex and complementary wind power from the regional grid. With a carbon intensity of just 18 gCO₂/kWh, this hub is among the cleanest GPU compute facilities in the world. The arid Saharan climate enables free-air cooling for over 8,500 hours per year, and the zero-water-consumption cooling architecture eliminates the water usage that plagues conventional data centers. A 15-year PPA with the Noor complex provides price stability at \$0.022/kWh.

Dakhla: Wind-Dominant Profile

The Dakhla hub operates at 94.8% renewable energy, powered predominantly by the exceptional Atlantic wind resource that delivers average speeds of 9.7 m/s. At 32 gCO₂/kWh, Dakhla's carbon intensity is the second-lowest in the Harch network. The wind-dominant profile provides a natural complement to Ouarzazate's solar generation, as Atlantic winds peak during evening and nighttime hours. Energy costs of \$0.018/kWh -- 72% below the EU average -- are secured through 15-year PPAs. Like Ouarzazate, Dakhla operates with zero water usage for cooling, relying on direct air and liquid cooling systems that require no evaporative water consumption.

Benguerir: Solar-Wind Hybrid

The Benguerir hub operates at 88.5% renewable energy with a carbon intensity of 55 gCO₂/kWh, drawing from a balanced mix of solar and wind generation. Located on the Mohammed VI Polytechnic University green campus, this hub benefits from the university's own renewable energy installations and serves as a pilot site for next-generation cooling technologies including direct-to-chip liquid cooling and immersion cooling. The hub operates with zero water usage for cooling and targets a PUE below 1.05 in future deployments with advanced cooling systems.

Tanger: Wind-Grid Mix

The Tanger hub operates at 82.1% renewable energy with a carbon intensity of 95 gCO₂/kWh, reflecting its position at the interface between renewable-rich and grid-supplied electricity. As the network gateway to Europe, Tanger's primary role is low-latency inference serving, where the carbon trade-off is offset by the

network performance requirements of real-time applications. Tanger employs closed-loop liquid cooling that minimizes water consumption, and the hub is transitioning to higher renewable fractions through additional wind PPA contracts expected to be finalized in 2026.

Casablanca: Urban Grid Mix

The Casablanca hub operates at 45% renewable energy with a carbon intensity of 210 gCO₂/kWh, reflecting its position within Morocco's largest urban grid. While this hub has the highest carbon intensity in the Harch network, it serves a critical function as the enterprise services hub, handling API gateway operations, control plane functions, and compliance-sensitive workloads that must remain within Moroccan jurisdiction. The hub is targeted for renewable energy improvement through planned rooftop solar installations and additional PPA procurement, with a target of 65% renewable by 2027.

Hub	Renewable %	gCO ₂ /kWh	Primary Source	PUE Target	Free Cooling (hrs/yr)	Water for Cooling
Ouarzazate	97.2%	18	Solar + Wind	<1.10	8,500+	Zero
Dakhla	94.8%	32	Wind (9.7 m/s)	<1.12	7,800+	Zero
Benguerir	88.5%	55	Solar + Wind	<1.08	8,200+	Zero
Tanger	82.1%	95	Wind + Grid	<1.15	6,500+	Minimal (closed-loop)
Casablanca	45%	210	Urban grid mix	<1.20	4,200+	Moderate (evaporative)

Table 4: Detailed carbon profile of each Harch GPU hub.

Hub	Solar (MW)	Wind (MW)	Grid (MW)	Battery (MWh)	PPA Duration
Ouarzazate	120 (Noor)	40	15	60	15 years
Dakhla	20	100 (Tarfaya)	10	50	15 years
Benguerir	50 (campus)	30	10	30	15 years
Tanger	10	60	30	35	15 years
Casablanca	5 (rooftop)	10	85	25	10 years

Table 5: Renewable energy sources and procurement structure per hub.

Life Cycle Carbon Assessment

Embodied Carbon

Embodied carbon -- the greenhouse gas emissions associated with manufacturing, transporting, and installing hardware -- represents a significant and often overlooked component of AI infrastructure's total carbon footprint. A single NVIDIA H100 GPU has an estimated embodied carbon footprint of approximately 30 kgCO₂e, while a full server node containing eight GPUs, along with CPUs, memory, storage, networking, and power supplies, carries an embodied carbon burden of approximately 800-1,000 kgCO₂e. Across Harch's fleet of 1,798 GPUs, total embodied carbon is estimated at approximately 200-250 tonnes CO₂e, including servers, networking equipment, cooling infrastructure, and facility construction.

Harch addresses embodied carbon through three strategies. First, we maximize the productive lifespan of each GPU by deploying hardware in thermally optimized environments that reduce component stress, targeting a 5-7 year operational life versus the industry average of 3-4 years. Second, we prioritize hardware refurbishment and component-level repair over full server replacement, reducing the embodied carbon of fleet upgrades by approximately 40%. Third, we maintain a decommissioning pipeline that routes retired hardware to certified refurbishment partners, ensuring that functional components re-enter the supply chain rather than being sent to landfill.

Operational Carbon

Operational carbon -- the emissions from electricity consumed during the use phase -- is where Harch's advantage is most pronounced. At an average carbon intensity of approximately 47 gCO₂/kWh, the operational carbon of a GPU hour on the Harch Platform is approximately 89% lower than the industry average of 450 gCO₂/kWh. For a training workload consuming 700W per GPU over 1,000 GPU-hours, this translates to approximately 33 kgCO₂e on Harch versus approximately 315 kgCO₂e on a conventional cloud provider -- a difference of 282 kgCO₂e, or roughly the emissions of driving a passenger car 1,800 km.

The PUE of Harch's infrastructure further amplifies this advantage. With an average PUE below 1.15, compared to the industry average of 1.4-1.6, Harch converts a larger fraction of consumed electricity into useful computation rather than overhead such as cooling and power distribution losses. This means that even before accounting for the lower carbon intensity of the electricity itself, Harch requires approximately 20-30% less total energy per unit of compute than a typical hyperscaler data center.

End-of-Life and Circular Economy

Harch is committed to a circular economy approach for all hardware assets. At decommissioning, each piece of equipment undergoes a three-stage assessment: functional testing determines whether the component can be redeployed within the Harch network (for example, shifting inference-optimized GPUs to less demanding workloads); refurbishment assessment determines whether the component can be restored to functional condition and sold to secondary markets; and finally, materials recovery ensures that components reaching true

end-of-life are processed through certified e-waste recyclers who recover precious metals, rare earth elements, and other valuable materials. Our target is to divert 95% of decommissioned hardware from landfill by 2028, and we are on track to achieve this target with current diversion rates exceeding 87%.

Phase	Harch (tCO ₂ e/yr)	Industry Avg (tCO ₂ e/yr)	Harch Advantage	Key Driver
Embodied (hardware)	~50	~50	Similar	5-7 yr lifespan vs 3-4 yr
Operational (energy)	~180	~1,620	-89%	47 vs 450 gCO ₂ /kWh
Cooling overhead	~20	~65	-69%	PUE <1.15 vs 1.4-1.6
Network/transit	~15	~15	Similar	Submarine cable efficiency
End-of-life	~5	~15	-67%	95% diversion target
Total	~270	~1,765	-85%	Renewable energy + efficiency

Table 6: Life cycle carbon assessment comparison (estimated annual emissions for 1,798 GPU fleet).

Comparison with Industry

To contextualize Harch's sustainability performance, we present a direct comparison with the published carbon metrics of the three largest hyperscaler cloud providers and OVHcloud, the European cloud provider with the most ambitious sustainability program. All data is sourced from publicly available sustainability reports, environmental disclosure filings, and grid carbon intensity databases. Where exact figures are not published, we use the best available estimates from third-party research and the grid carbon intensity of the provider's primary European data center regions.

Metric	Harch	AWS	GCP	Azure	OVHcloud
Avg carbon intensity (gCO ₂ /kWh)	47	~316	~178	~334	~90
Renewable energy %	81.5%	~42%	~56%	~38%	~80%
PUE (average)	<1.15	1.15-1.35	1.10-1.20	1.20-1.40	1.20-1.30
Water-free cooling	3 of 5 hubs	Limited	Limited	No	Partial
Carbon-aware scheduling	Native (HarchOS)	No	Pilot	No	No
Free cooling hours/yr	~8,500	~2,000	~3,000	~2,500	~4,000
PPA duration (years)	15	10-12	10-15	10-12	10-15
Battery storage (MWh)	200	Not disclosed	Not disclosed	Not disclosed	Limited

Table 7: Sustainability metrics comparison across major cloud and AI infrastructure providers.

The comparison reveals that Harch achieves the lowest operational carbon intensity of any major AI infrastructure provider, at approximately 47 gCO₂/kWh versus 90 gCO₂/kWh for OVHcloud and 178-334 gCO₂/kWh for the US hyperscalers. Harch is also the only provider that offers native carbon-aware scheduling as a built-in platform capability, rather than as an experimental feature or manual configuration option. The combination of Morocco's renewable energy resources, purpose-built cooling architecture, and intelligent workload scheduling creates a sustainability advantage that cannot be replicated by retrofitting existing infrastructure in less favorable geographic locations.

It is worth noting that the US hyperscalers have made substantial commitments to renewable energy procurement, and their reported renewable energy percentages include significant quantities of Renewable Energy Certificates (RECs) and power purchase agreements that may not reflect the actual carbon intensity of the electricity consumed by their data centers at any given hour. Harch's renewable energy figures reflect the actual generation mix at each hub location, verified by real-time grid monitoring data, providing a more accurate and auditable representation of the true carbon footprint of compute workloads.

Water Usage & Cooling

Water-Free Cooling Architecture

Water consumption is an increasingly critical dimension of data center sustainability, particularly in water-stressed regions where large-scale evaporative cooling can compete with agricultural and municipal water needs. The Harch Platform addresses this challenge head-on: three of our five hubs -- Ouarzazate, Dakhla, and Benguerir -- operate with zero water usage for cooling. This is achieved through a combination of direct air-side free cooling, which exploits the region's average ambient temperature of approximately 22 degrees Celsius to provide cooling without mechanical refrigeration for over 8,500 hours per year, and closed-loop liquid cooling systems that transfer heat from GPU components to external dry coolers without evaporative water loss.

The Tanger hub employs closed-loop liquid cooling with minimal water usage, limited to occasional makeup water for the closed-loop system. The Casablanca hub, as the most recently commissioned facility, currently uses a hybrid cooling system that includes some evaporative cooling, but a transition plan is underway to convert this hub to a fully water-free architecture by 2027. Our Water Usage Effectiveness (WUE) target is 0 L/kWh across all hubs by 2028, and current performance is on track to achieve this goal.

Liquid Cooling Architecture

As GPU thermal design power (TDP) continues to increase -- the NVIDIA H100 draws 700W per GPU, and next-generation accelerators are expected to exceed 1,000W -- air cooling alone becomes insufficient for maintaining optimal operating temperatures. Harch's liquid cooling architecture employs direct-to-chip cold plates that circulate coolant directly over GPU and CPU heat spreaders, achieving heat transfer coefficients 3-5x higher than air-based cooling. This approach not only eliminates the need for water-intensive evaporative cooling but also reduces the energy required for heat rejection, contributing to the platform's industry-leading PUE of below 1.15. The Benguerir hub is currently piloting full immersion cooling, where servers are submerged in a dielectric fluid that provides even more efficient heat removal, with the potential to reduce PUE below 1.05.

Hub	Primary Cooling	Water Usage	Avg Temp	Free Cooling hrs/yr	PUE
Ouarzazate	Air + Liquid (closed-loop)	Zero	~22°C	8,500+	<1.10
Dakhla	Air + Liquid (closed-loop)	Zero	~21°C	7,800+	<1.12
Benguerir	Liquid + Immersion (pilot)	Zero	~23°C	8,200+	<1.08
Tanger	Liquid (closed-loop)	Minimal	~20°C	6,500+	<1.15
Casablanca	Hybrid (transitioning)	Moderate	~19°C	4,200+	<1.20

Table 8: Cooling architecture and water usage profile per hub.

Our Pledge & Targets

Harch Corp commits to transparent, measurable, and independently verifiable sustainability targets. Our pledges are not aspirational statements; they are binding commitments with defined timelines, quantitative metrics, and public reporting obligations. We believe that the credibility of sustainability claims depends on the rigor with which they are measured and the transparency with which they are reported. To that end, all targets listed below will be reported against in our annual sustainability report, with progress verified by an independent third-party auditor.

2027 Targets

- Achieve 90% average renewable energy across all hubs (currently 81.5%)
- Reduce average carbon intensity to below 35 gCO₂/kWh (currently ~47 gCO₂/kWh)
- Convert Casablanca hub to water-free cooling architecture
- Increase battery storage capacity to 400 MWh (currently 200 MWh)
- Achieve PUE below 1.10 across all hubs (currently <1.15 average)

2030 Targets

- Achieve net-zero operational carbon emissions across all hubs
- Reach 95% average renewable energy penetration
- Achieve zero waste-to-landfill for all decommissioned hardware (95% target by 2028)
- Reduce embodied carbon per GPU-hour by 50% through extended hardware lifespan and refurbishment
- Publish real-time carbon intensity data for all hubs via public API

Transparency Commitments

Harch commits to the following transparency practices: publication of an annual sustainability report with independently verified data; real-time publication of grid carbon intensity data at each hub through a public API endpoint; quarterly disclosure of renewable energy percentages and PUE metrics; inclusion of Scope 1, 2, and 3 emissions in all carbon reporting; and adoption of the GHG Protocol and SBTi methodologies for emissions accounting. We invite our customers, partners, and critics to hold us accountable against these commitments.

Target	Current (2025)	2027 Goal	2030 Goal
Renewable energy %	81.5%	90%	95%
Avg carbon intensity (gCO ₂ /kWh)	~47	<35	<20 (net-zero ops)
PUE (average)	<1.15	<1.10	<1.08
Water-free cooling hubs	3 of 5	5 of 5	5 of 5

Battery storage (MWh)	200	400	600
Hardware diversion from landfill	87%	92%	95%+
Real-time carbon API	Internal	Public beta	Full public

Table 9: Harch sustainability targets roadmap, 2025-2030.

Get Involved

Carbon Footprint Calculator

Harch provides a free, open-source carbon footprint calculator that enables organizations to estimate the carbon emissions of their AI workloads on different infrastructure providers. The calculator accepts parameters including the number and type of GPUs, expected utilization hours, the geographic region of the compute infrastructure, and the carbon intensity of the local grid. It then produces a detailed breakdown of operational carbon emissions, cooling overhead, and estimated embodied carbon allocation, enabling like-for-like comparisons between Harch and alternative providers. The calculator is available at calculator.harch.ai and its source code is published on GitHub under an MIT license, ensuring full transparency of the methodology and assumptions.

Partnership Opportunities

Harch actively seeks partnerships with organizations that share our commitment to sustainable AI infrastructure. We offer several collaboration models. Research partnerships provide academic institutions with subsidized access to Harch GPU capacity for sustainability-focused AI research, including model efficiency, carbon-aware scheduling algorithms, and green AI benchmarking. Technology partnerships enable hardware vendors, cooling system manufacturers, and renewable energy developers to test and validate their products within Harch's operational environment, with real-world performance data and carbon metrics. Corporate sustainability partnerships allow enterprises to integrate Harch's carbon reporting directly into their own ESG disclosures, with automated Scope 2 and Scope 3 emissions calculations based on actual workload data rather than industry averages.

Contact

For inquiries related to sustainability reporting, carbon footprint assessments, partnership opportunities, or access to our real-time carbon intensity API, please contact the Harch Sustainability Team. We are committed to responding to all inquiries within 48 business hours and to providing the data and documentation necessary for informed decision-making.

Sustainability Team: sustainability@harch.ai

Carbon API: api.harch.ai/carbon

Calculator: calculator.harch.ai

GitHub: github.com/harch-corp

General Inquiries: info@harch.ai

This report was prepared by the Harch Corp Sustainability Team. All data presented herein is based on verified operational metrics and third-party grid carbon intensity data as of Q1 2025. Methodology follows the GHG Protocol Corporate Standard and SBTi guidance. Independent third-party verification of all metrics is planned for Q3 2025.

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